1. Experiment in short

Number of nodes	4
Number of topologies	3
Location of the experiment	Sesimbra, Portugal, $(38.26105°N, 8.86776°W)$
Measurement duration	about 1 hour
Measurement granularity	1 second
Link quality measurements	On/Off status

2. Location

The experiment took place on July 22, 2016, in the Atlantic Ocean in the proximity of Sesimbra, south of Lisbon, Portugal. The Google map picture in Figure 1 points to the experiment's location via a round yellow *ship* marker. The experiment involved 4 nodes: 2 nodes were lowered from the NATO Research Vessel (NRV) Alliance, 1 node was deployed from a working boat (RHIB) and one last node was attached to a moored Gateway Buoy (GW) in the area. The average depth at which the modems were deployed was 10 meters. The experiment took place the 22nd of July and comprised 3 topologies: the session started at 10:40 AM and lasted until around 11:42 AM local time.

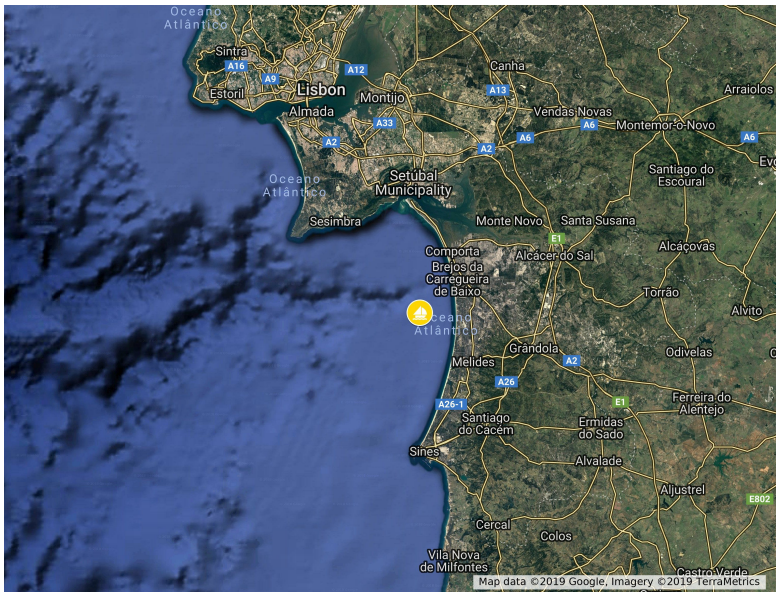
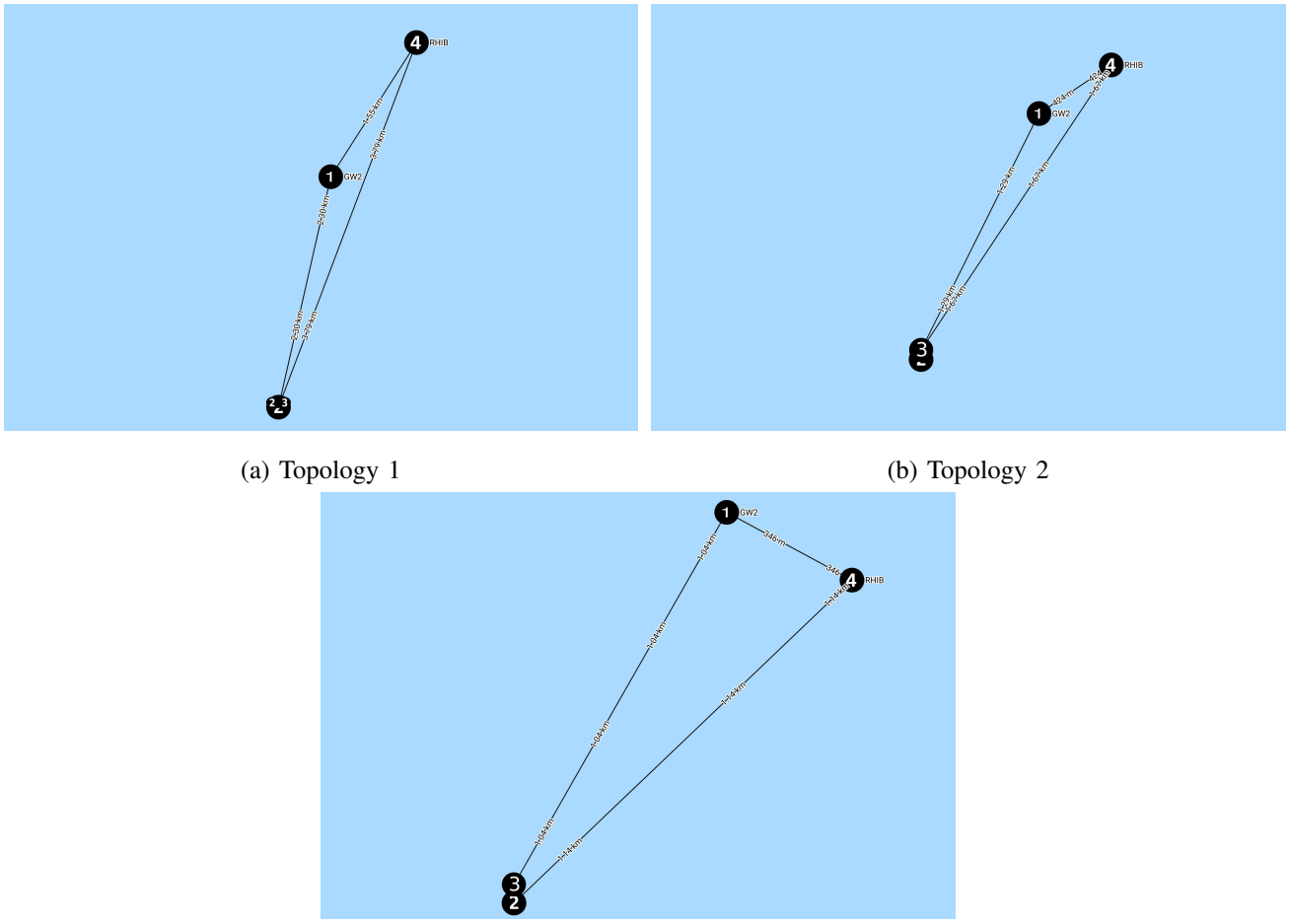


Fig. 1: Satellite picture of the sea trial location (image from Google maps).



(a) Topology 1

(b) Topology 2

(c) Topology 3

Fig. 2: The 3 tested topologies

3. Topologies

Fig. 2 shows the initial location of the nodes in each of the three topologies. Except the node attached to the Gateway Buoy, all nodes were drifting throughout the experiment. In particular, the node deployed from the working boat (RHIB) was subject to larger drifts than the nodes deployed from the NRV Alliance.

The 2 nodes deployed from the NRV Alliance were placed at the stern and at the bow so that their separation was, approximately, 60 m. The node onboard the GW is considered to be located at a fixed in position.

4. Description of the nodes

All nodes communicated via EvoLogics SC2R 18/34 modems, operating in the 18-34 kHz frequency band using EvoLogics's proprietary sweep-spread carrier (S2C) modulation scheme. For further information, the interested reader is referred to EvoLogics's web site: https://www.evologics.de/en/products/acoustics/index.html.

The transmissions exploited the *low-level firmware* of the devices as explained in [2]: this feature allowed us to bypass the proprietary DMAC MAC-level protocol installed by default on the EvoLogics devices.

5. Tested protocols and schemes

The experiment was conceived to test the emergency communication protocol presented in [1]. The protocol operated as follows:

1) A node, in this case the Gateway Buoy (GW), sent out a message that is clearly identified, by the other nodes, as an SOS message; in this particular case a long sequence of a repeated short message was sent: this first transmitter is considered to be the source of the SOS;
2) each other node that received the SOS message encoded that message plus a timestamp reference (in this case the *epoch* of the running machine) and sent out its own message;
3) each other subsequent receiving node is now able to discriminate if the source is the GW (SOS message) or another node (encoded message).

The above process served as a data collection tool to provide information to topology estimation techniques, with the objective of localizing the SOS source.

6. Link Measures

The network links are evaluated by means of on/off measurements, which indicate whether a packet transmitted at a given time instant would be received or not.

7. Data Available

The stored data is given in the *REP-07-16.mat* file. This file brings together 4033 measurement epochs, which have been taken at constant intervals of 1 second starting from the beginning of each experiment. The `.mat` file includes all topologies presented in Fig. 1 and contains four types of matrices:

- the location matrices `LocMat1` to `LocMat3`, of size $1640\times4\times3$, $1632\times4\times3$, and $761\times4\times3$;
- the topology measurement matrices `TopMat1` to `TopMat3`, of size of size $1640\times4\times3\times1$, $1632\times4\times3\times1$, and $761\times4\times3\times1$;
- the adjacency matrices `AdjMat1` to `AdjMat3` of size $1640\times4\times4$, $1632\times4\times4$, and $761\times4\times4$;
- the technology matrices `TechMat1` and `TechMat2` of size $1640\times4\times1$, $1632\times4\times1$, and $761\times4\times1$;.

Additionally, for each matrix type, we concatenate all matrices into a single matrix, respectively named:

- `FullLocMat` of size $4033\times4\times3$;
- `FullTopMat` of size $4033\times4\times4\times1$;
- `FullAdjMat` of size $4033\times4\times4$;
- `FullTechMat` of size $4033\times4\times1$.

Using Matlab/Octave matrix notation:

- `LocMat1(t,i,:)` to `LocMat3(t,i,:)` or `FullLocMat(t,i,:)` are the UTM coordinates (zone 29S) and the depth of node i at time instant t;
- `TopMat1(t,i,j,p)` to `TopMat3(t,i,j,p)` or `FullTopMat(t,i,j,p)` is the measured on/off status for technology p over the link $i \rightarrow j$ at time t;
- `AdjMat1(t,i,j)` to `AdjMat3(t,i,j)` or `FullAdjMat(t,i,j)` is equal to 1 if there exists a topology link between nodes i and j at time t;
- `TechMat1(t,i,p)` to `TechMat3(t,i,p)` or `FullTech(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. References to detailed description

Further details about this experiment are given in the following papers:

[1] R. Diamant, R. Francescon, M. Zorzi, "A graph localization approach to assist a diver-in-distress," *Proc. WPNC*, Bremen, Germany, Oct. 2017.

[2] F. Campagnaro, R. Francescon, O. Kebkal, P. Casari, K. Kebkal, M. Zorzi, "Full Reconfiguration of Underwater Acoustic Networks through Low-Level Physical Layer Access," *Proc. ACM WUWNet*, Halifax, Canada, Nov. 2017.